Application of Empirical Mode Decomposition Combined With Notch Filtering for Interpretation of Surface Electromyograms During Functional Electrical Stimulation

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***Abstract*— The goal of this paper is to demonstrate a novel approach that combines Empirical Mode Decomposition (EMD) with Notch filtering to remove the electrical stimulation (ES) artifact from surface electromyogram (EMG) data for interpretation of muscle responses during functional electrical stimulation (FES) experiments. FES was applied to the rectus femoris (RF) muscle unilaterally of six able bodied (AB) and one individual with spinal cord injury (SCI). Each trial consisted of three repetitions of ES. We hypothesized that the EMD algorithm provides a suitable platform for decomposing the EMG signal into physically meaningful intrinsic mode functions (IMFs) which can be further used to isolate electrical stimulation (ES) artifact. A basic EMD algorithm was used to decompose the EMG signals collected during FES into IMFs for each repetition separately. IMFs most contaminated by ES were identified based on the standard deviation (SD) of each IMF. Each artifact IMF was Notch filtered to filter ES harmonics and added to remaining IMFs containing pure EMG data to get a version of a filtered EMG signal. Of all such versions of filtered signals generated from each artifact IMF, the one with maximum signal to noise ratio (SNR) was chosen as the final output. The validity of the filtered signal was assessed by quantitative metrics, 1) root mean squared error (RMSE) and signal to noise (SNR) ratio values obtained by comparing a clean EMG and EMD-Notch filtered signal from the combination of simulated ES and clean EMG and, 2) using EMG-force correlation analysis on the data collected from AB individuals. Finally, the potential applicability of this algorithm on a neurologically impaired population was shown by applying the algorithm on EMG data collected from an individual with SCI. EMD combined with Notch filtering successfully extracted the EMG signal buried under ES artifact. Filtering performance was validated by smaller RMSE values and greater SNR post filtering. The amplitude values of the filtered EMG signal were seen to be consistent for three repetitions of ES and there was no significant difference among the repetition for all subjects. For the individual with a SCI the algorithm was shown to successfully isolate the underlying bursts of muscle activations during FES. The data driven nature of EMD algorithm and its ability to act as a filter bank at different bandwidths make this method extremely suitable for dissecting ES induced EMG into IMFs. Such IMFs clearly show the presence of ES artifact at different intensities as well as pure artifact free EMG. This allows the application of Notch filters to IMFs containing ES artifact to further isolate the EMG. As a result of such stepwise approach, the extraction of EMG is achieved with minimal data loss. This study provides a unique approach to dissect and interpret the EMG signal during FES applications.**

***Index Terms*— Artifact removal, electromyogram (EMG) filtering, empirical mode decomposition, functional electrical stimulations.**

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I. Introduction

TRADITIONALLY, electrical stimulation (ES) has been used to assist or restore neuromuscular function to paralyzed muscle and potentially ameliorate many of the secondary consequences associated with the neurological injury [1]–[9]. Functional Electrical Stimulation (FES) operates on the basic principle that the application of electrical current to a nerve can elicit action potentials in peripheral neurons disconnected from voluntary control [5]. The strength of the resultant muscle contraction can be regulated by modulating the stimulus parameters [1], [5]. Clinical and research interventions using chronic application of ES have been shown to improve tissue health or voluntary function by inducing physiological changes that potentially remain beyond the stimulation period [1], [2], [4]–[6], [8], [9]. Further, functional inter limb movement can be produced by coordinating stimulation of multiple paralyzed muscles, providing the basis for various commercially available FES neuroprosthetic devices [3], [5], [10], [11].

ES therapeutic intervention has tremendous potential [4], [5], [11], however due to the lack of sensitive measurement tools to dissociate the stimulation from the underlying

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activity, the direct or real time effect of ES on the muscle or nerve cannot be measured, rather the effect of stimulation on functional performance is often determined by mechanical output or a clinical performance measure [12], [13]. Neither of these assessment tools directly measure the effect of stimulation on the specific muscle being stimulated.

Currently, the neuromuscular or neurophysiological effects of ES have been assessed through the collection of surface electromyograms (EMG), however EMG during ES has proven to be difficult to interpret [14]–[16] due to the presence of stimulus artifact. The stimulus artifact is a broad band signal with wide-spread stimulus frequency harmonics at high amplitudes that completely engulfs the EMG. Removal of the stimulus artifact presents a signal processing challenge due to the overlapping power spectra of the ES stimulus and EMG signal, and standard filtering methods result in uninterpretable signal distortion [17].

Historically, researchers have modified different hardware [14], [16], [18] and software [19]–[21] approaches to elicit the EMG signal albeit each approach has limitations. Hardware modifications have included suppressing the stimulus artifact "online" during signal acquisition utilizing a sample-and-hold technique [14]. In this technique, the circuit is switched to "hold" mode immediately prior to stimulation, thereby stopping the recording amplifier from detecting the stimulus artifact. At the end of the "hold" period, the circuit is switched back to "sample" mode and the neural signal is again passed through to the recording amplifier [14]. Though effective for suppressing the stimulus artifact, this technique provides limited information about the muscle contraction at the precise moment of stimulation.

An "offline" software based approach to remove ES artifact from EMG signals is offered by O'Keeffe et al. using a two stage peak-detection algorithm where a threshold value will detect the artifact peaks followed by the removal of the artifact by replacing the data with zeros for the artifact duration [15]. However, this algorithm is not reliable as the M-wave and the artifact spike overlap. A second approach developed by Heffer and Fallon, is to remove the artifact by replacing the sample points at each stimulus artifact event with values interpolated along a straight line, computed from neighboring sample points [17]. This technique requires an identifiable artifact event and the artifact duration must remain less than both the inter-stimulus interval and the time course of the action potential [17]. More recently, Qiu et al. developed more effective yet computationally intense approach to remove the artifact during FES using genetic algorithm based adaptive match filter (AMF) [21]. Considering the constraints or selective advantages of present techniques for artifact reduction, there is a definite need for a robust algorithm to clearly separate out the stimulus signal from the EMG, without the significant need for prior information about the stimulus. Optimally, the algorithm should operate effectively with or without the temporal overlap between the electrical stimulus and EMG signal.

Huang et al. have introduced a novel approach for analyzing non-stationary and non-linear data called as Empirical Mode Decomposition (EMD) [22]. This algorithm decomposes the signal based on the direct extraction of the energy associated with various intrinsic time scales [22], providing the opportunity to dissect non-stationary signals such as EMG to thoroughly understand the underlying phenomenon [20], [23]–[31]. Further, Huang et al proposed the EMD– Hilbert transform (HT) as more appropriate tool for time-frequency analysis (TFA) for analyzing a non-stationary signal as compared to Fourier based TFA tools which assume stationarity of a signal [22]. The time frequency plot obtained using the EMD-HT approach provides a higher resolution than any traditional Fourier and wavelet based time-frequency distribution thus allowing a better way of analyzing a signal in the time-frequency domain, especially if the signal under investigation is non-linear and non-stationary [22], [32], [33]. Recently, EMD algorithm has been successfully applied for the removal of tissue artifact from respiratory signals [29] and a similar adaptation of this approach removed respiratory, motion, and cardiac artifacts from electrogastrogram (EGG) signals [27]. Previously we have provided evidence that EMD could prove to be a suitable platform for the decomposition of a non-stationary EMG signal into intrinsic mode functions (IMFs), making it possible to observe the instantaneous characteristics with better resolution than the traditional methods for isolation of the EMG signal from ES artifact [20]. However, high frequency components of the EMG data were lost in discarded IMFs during the process of reconstruction as EMD acts as a filter-bank and extracts the IMFs of frequencies from high to low order [22], [34]. This suggests that the basic EMD extracts low frequency components of muscle activations from EMG signal buried under ES artifact, but requires further filtering of discarded IMFs to recover the high frequency components for accurate extraction of muscle responses. Further, EMD based decomposition allows the distribution of ES intensity throughout multiple IMFs, which presents the opportunity to apply standard frequency selective band-stop or Notch filtering on the IMF containing the low intensity ES artifact and further recover the EMG components that were lost during previous approach [20]. Therefore, the primary purpose of this paper is to demonstrate the utility of EMD combined with Notch filtering to extract EMG during ES with minimal loss of data and higher signal to noise (SNR) ratio.

II. Methods

A. Subjects

Six able bodied (AB) (Age: 30.67±14.02 years, height: 179.92± 8.71 cm, weight: 81.59± 3.98 kg) and 1 person with SCI (Age: 26 years, height: 162.56, weight: 90.91 kg) were recruited for this study (see Table I).

The individual with SCI was classified as having an American Spinal Injury Association Impairment Scale (AIS) B, a T6 neurological level of injury. Inclusion criteria for all subjects included: 1) must be able to tolerate ES and show visible contraction; 2) must be able to remain in an upright position for 30 min; 3) no history of femur or tibia fracture, extensive bone trauma, history of prior bone disease; 4) No presence of major cardiovascular or pulmonary disease, marked obesity, or lower extremity deep vein thrombosis; 5) no skin

TABLE I
STUDY PARTICIPANT DEMOGRAPHICS AND ES PARAMETERS

Subject	*Type*	*Age (y)*	*Height (cm)*	*Weight (kg)*	ES_{min} *(mA)*	ES_{max} *(mA)*
1	AB	26	182.9	84.1	13	37
2	AB	24	182.9	77.3	7	39
3	AB	25	190.5	81.8	7	31
4	AB	59	182.9	86.4	14	58
5	AB	28	165.1	83.6	11	47
6	AB	22	175.3	76.4	8	46
Mean		**30.67**	**179.92**	**81.59**	**10**	**43**
SD		**14.02**	**8.71**	**3.98**	**3.1**	**9.45**
1	SCI	26	162.6	90.9	21	91

lesions or skin breakdown, major gastrointestinal problems or conductive heart problems. All the subjects signed a consent form approved by the Kessler Foundation Institutional Review Board prior to study participation.

B. Testing Procedures

All participants were positioned in a standing frame Easy-Stand Evolv (Altimate Medical, Morton, MN, USA), secured against movement and were standing with their knees flexed at 20° as measured by a goniometer. ES was applied to the rectus femoris (RF) muscle unilaterally. The ES electrodes were placed at the medial and distal ends of right RF (RRF), where the subject achieved maximum contractions. ES was delivered via an EMPi 300PV FES unit (Empi Inc., St. Paul, MN, USA) using symmetrical biphasic pulses of 300 μs at 35 Hz applied across a 1000 Ω load. Surface EMG (Noraxon, Scottsdale, AZ, USA) electrodes were placed midpoint between the two ES electrodes and also on the lateral RF to collect ES induced RF muscle activation. EMG data were also collected from RF and lateral RF of the non-stimulated limb. The force produced by ES induced contraction of the RF while standing was measured by a 6 degrees of freedom load cell (AMTI, AMTI. Watertown, MA, USA) located in the anterior kneepad of a standing frame. All analog data (EMG and Forces) were collected at 2520 Hz.

The trial was conducted as follows. The intensity for the ES began at an amplitude 6mA below the minimal threshold amplitude, ES_{min} for the first visible contraction of muscle. Starting from ES_{min}, the stimulation was increased by 2 mA every 4 s up to ES_{max}. ES_{max} represents the maximal tolerable ES intensity level for a subject (Table I). ES was kept constant for 6 s after reaching ES_{max} and then was decreased to zero. The initial repetition within the trial was repeated two times so there was a total of three repetitions with a rest period of approximately 1 min between each repetition [35].

C. Data Processing

EMD-Notch Filtering: EMG data were band pass filtered between 20 to 150 Hz prior to EMD-Notch filtering. Fig. 1 describes the filtering scheme implemented for extraction of ES artifact from the EMG data. Each stage of the filtering operation is described below.

Stage I: Empirical Mode Decomposition (EMD)

The process called *Sifting* is used to identify intrinsic oscillatory modes in a signal x(t) and when repeatedly applied,

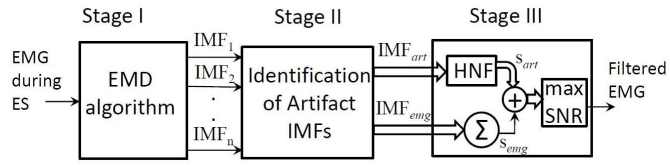

Fig. 1. Filtering scheme to remove ES artifact from RRF EMG signal. HNF: harmonic Notch filter; SNR: signal to noise ratio.

the signal is decomposed into IMFs (and a residual) [22]. The steps involved in this decomposition are shown below.

1) Signal local maxima and minima were identified.
2) The local maxima were connected by an envelope obtained by fitting cubic spline. The process is repeated for local minima.
3) The mean of the maxima envelope and minima envelope, m_1 gives the first component which is then subtracted from $x(t)$ to give h_1
4) Next, h_1 is inspected to see if it is an IMF. For a function to be called as IMF, the number of extrema and number of zero crossings must be equal or differ at most by one and the mean of the envelope defined by the local maxima and the one defined by local minima should be zero [22]. Upon satisfying these conditions, h_1 was considered as an IMF. If not h_1 was treated as a new data to get h_{11} using step 1 to 3. The sifting process is repeated until h_{1k} is an IMF,

$$h_{1(k-1)} - m_{1k} = h_{1k} = c_1 \tag{1}$$

5) Separate c_1 from the original signal and repeat the process on the residual until no further IMFs can be retrieved. What remains is the last residual r_n. Hence,

$$x(t) = \sum_{i=1}^{n} c_i + r_n. \tag{2}$$

Excessive application of the sifting process can result in meaningless IMFs. To avoid this, a limitation on the standard deviation calculated (SD> 0.1) from two consecutive sifting processes is used as a stopping criterion for sifting [22]. The final result of the sifting process is the decomposition of the signal x(t) into n IMFs with final residual r_n.

Stage II: Identification of Artifact IMFs

IMFs generated during Stage I are assessed for their standard deviations (SD). Let $S_d = [SD_1\ SD_2 \ldots SD_n]$ where $SD_i = std(\mathrm{IMF}_i)$. The ES artifact is generally higher in amplitudes compared to the actual muscle response by a factor of 3 [36] and this difference in the amplitudes could be utilized to separate the artifact and M-wave components [37]. With this rationale, the threshold, Th_{sd} was set as,

$$Th_{sd} = 3M_d, \text{where} M_d = median(S_d) \tag{3}$$

IMFs with standard deviations greater than Th_{sd} are collectively called IMF_{art} and are considered to be the most affected by ES artifact. Remaining IMFs are collectively called IMF_{emg}.

$$\begin{aligned} \mathrm{IMF}_{art} &= [\mathrm{IMF}_1 \mathrm{IMF}_2 \ldots \mathrm{IMF}_m] \\ \mathrm{IMF}_{emg} &= [\mathrm{IMF}_{m+1} \mathrm{IMF}_{m+2} \ldots \mathrm{IMF}_n] \end{aligned} \tag{4}$$

Stage III: Application of Notch Filtering and final output

Typically, an EMG signal collected during FES contains ES stimulation artifact of 35 Hz and its 10 to 12 harmonics. Hence, a series of Notch filters (a total of 12 second-order Butterworth band-stop filters with cut off frequencies at 35 Hz and its subsequent harmonics) is applied on each IMF in IMF_{art} to filter out harmonics of ES frequency (35 Hz) to get S_{art}. Filter order of 2 was found to be sufficient to filter out 35 Hz and its harmonics (Fig. 6).

$$S_{art_i} = notch(IMF_{art_i}) \tag{5}$$

All the IMFs in IMF_{emg} containing pure EMG are added to get S_{emg}.

$$S_{emg} = \sum IMF_{emg} = \sum_{k=m+1}^{n} \text{IMF}_k \tag{6}$$

Next, each component in S_{art} is added to S_{emg} to produce $'m'$ versions of filtered signals.

$$y_i = S_{emg} + S_{art_i} \tag{7}$$

The version that gives maximum SNR is chosen as the final output, y as follows:

$$imax = \underset{i=1 to\, m}{\text{argmax}}\,[SNR(y_i)] \tag{8}$$

$$y = y_{imax}. \tag{9}$$

Note that the SNR in (8) was calculated by comparing each filtered version with the original ES contaminated EMG signal using generic form of (12a) and (12b).

Time Frequency Representation Using EMD-Hilbert Transform (EMD-HT): Once all the IMFs and a final residual were extracted from the data, a Hilbert transform (HT) was applied to represent each IMF in the following analytical form:

$$HT\{IMF\} = a(t)e^{-i\int \omega t.dt} \tag{10}$$

where a and ω represent the instantaneous energy and frequency for each IMF giving a time frequency representation in the form of (t, ω, a). Each IMF was plotted with a different color code depending on its energy value (a). The details of this method are reported in [22].

Force Data Processing: Force data was low pass filtered by 2-Hz Butterworth filter to clean high frequency noise in the signal.

Performance Evaluation of EMD-Notch Filtering Algorithm: For the validation of the EMD-Notch algorithm the following data collection and analyses was performed for one AB. A clean volitional EMG was collected [Fig. 8(a)] from the RF muscle of one AB individual standing prior to the ES trial. The subject was asked to produce volitional contraction of RF and perform knee extension during standing. Computationally, a simulated ES signal was generated at different amplitude ratios (4, 8, 12, 16, and 20) of ES to clean volitional EMG. The rationale for selecting higher amplitudes of simulated ES is that the EMG data collected during real FES applications often consists of ES artifact of much higher amplitudes compared to that of actual muscle activations. The simulated ES had the same profile (pulse repetition frequency, pulse width) [Fig. 8(b)] of the experimental ES. Simulated ES [Fig. 8(b)] and volitional EMG [Fig. 8(a)] were summed to produce a test signal that will be used to test the performance of the algorithm at different levels of the simulated ES intensities [Fig. 8(c)].

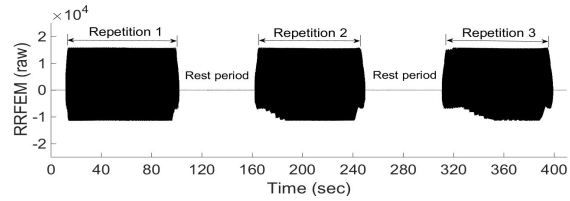

Fig. 2. Surface EMG collected from RRF during three repetitions of ES. The EMD-Notch algorithm was applied on each repetition separately.

Normalized root-mean-squared error (NRMSE) was calculated to quantify the accuracy of the method. $NRMSE_{pre}$ values were calculated by comparing the original volitional unstimulated EMG, $v(t)$, with simulated ES induced EMG, $e(t)$. $NRMSE_{post}$ values were calculated by comparing clean EMG, $v(t)$, with EMD-Notch filtered signal, $y(t)$, at different amplitudes of simulated ES, each with "n" samples (11a), (11b). Similarly, SNR values were calculated pre- and post- filtering (12a), (12b)

$$NRMSE_{pre} = \frac{\sqrt{\sum (e(t) - v(t))^2/n}}{SD(v\,(t))} \tag{11a}$$

$$NRMSE_{post} = \frac{\sqrt{\sum (y(t) - v(t))^2/n}}{SD(v\,(t))} \tag{11b}$$

$$SNR_{pre}(dB) = 10\log_{10}\frac{\sum v(t)^2}{\sum (e(t) - v(t))^2} \tag{12a}$$

$$SNR_{post}(dB) = 10\log_{10}\frac{\sum v(t)^2}{\sum (y(t) - v(t))^2}. \tag{12b}$$

D. Statistical Analysis

Analysis of Variance (ANOVA) with repeated measures were computed to distinguish the differences among the RMS EMG amplitudes during the three repetitions within the single trial as well as during different phases (P1, P2, and P3) of each repetition where, P1: ES_{min} to ES_{max} (ramping up); P2: ES_{max} plateau; P3: ES_{max} to zero (ramping down). Cross correlation (R_{xy}) values for the force waveform and EMD-Notch filtered EMG during each repetition were also compared using ANOVA with repeated measures.

III. Results

The EMG signal collected during each of the three repetitions of ES induced muscle contraction of the RF was used as a test signal (Fig. 2). The EMD-Notch algorithm was applied to each of the three repetitions separately. For the purpose of demonstration, the results showing the implementation of EMD-Notch algorithm on Repetition 1 are provided.

The basic EMD algorithm decomposed the signal into 16 IMFs as shown in Fig. 3. As mentioned in the EMD literature, the addition of all IMFs and the residual will yield the original signal with minimum error. By visual inspection of

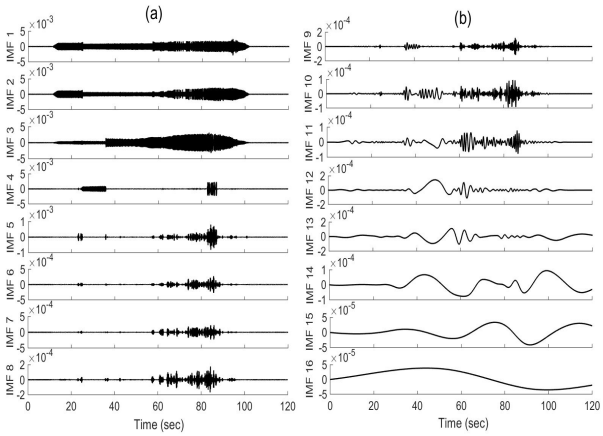

Fig. 3. Stage I - The decomposition of "Repetition 1" EMG signal performed using basic EMD method to give (a) IMFs 1 to 8, (b) IMFs 9 to 16.

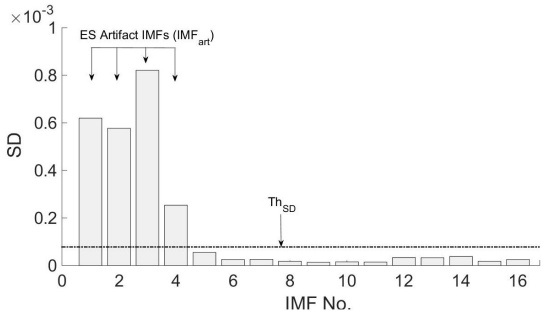

Fig. 4. Stage II—Identification of artifact modes based on standard deviations.

the data (Fig. 3), the stimulus artifact dominated the first four IMFs. IMFs 5 to 11 appear as smoothed versions of the EMG signal. These EMG waveforms, originally buried within the stimulus artifact, are now clearly visible after the application of EMD algorithm. From IMF 5 onwards, each mode consists of various bands of frequencies (high to low order) that constitute the EMG signal.

Fig. 4 demonstrates the outcome of Stage II. The SD values for IMFs are plotted. IMFs 1 to 4, each being an artifact mode, show greater SD values compared to the remaining modes (IMF 5 onwards) due to the presence of high amplitude ES artifact. The horizontal dotted line represents the threshold value calculated using (3), which was used to classify the IMFs into IMF$_{art}$ and IMF$_{emg}$. S_{emg} was then calculated as the sum of all IMF$_{emg}$ (IMF 5 to 16). Finally, at Stage III—a series of Notch filters $[f_{cutoff}(Hz) = [35, 70, 105, \ldots 280]]$ were applied on the IMF$_{art}$ with the smallest SD. The IMF with smallest SD contained the lowest degree of ES artifact, as well as high frequency EMG data. Output of the notch filtering, S_{art} was then added to S_{emg} to get the filtered signal $y(t)$ (5)–(7).

As seen in Fig. 5, the EMD algorithm was able to reconstruct the EMG signal. For better readability, the reconstructed signal is shown during a window of 10 s (75–85 s) to show bursts of ES induced muscle activations. The Fourier transforms of the original and the reconstructed EMG signal shows that the EMD-Notch algorithm was able to remove

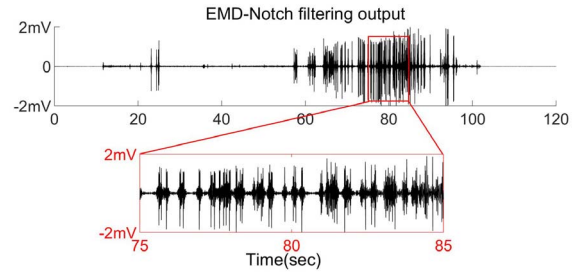

Fig. 5. Filtered EMG using EMD-Notch algorithm.

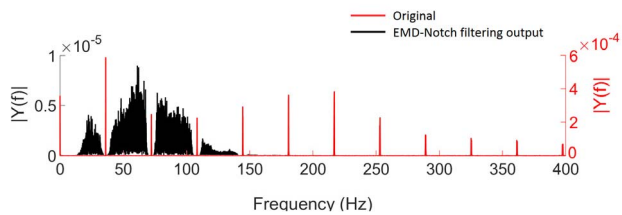

Fig. 6. Fourier transforms of original ES contaminated EMG and EMD-Notch filtering output for Repetition 1.

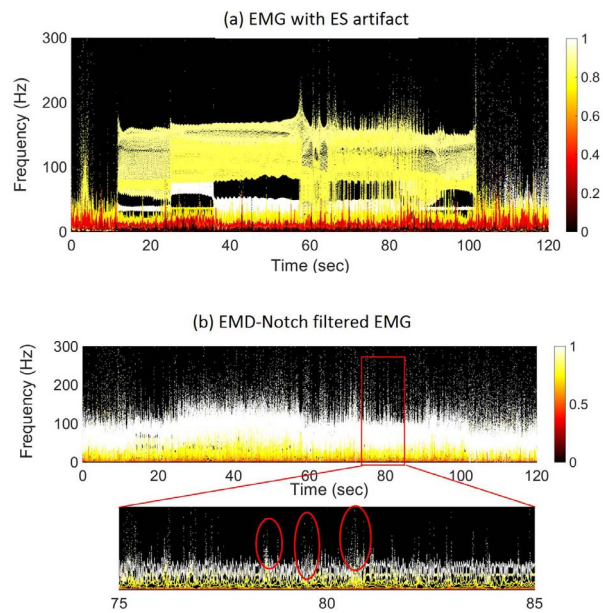

Fig. 7. Time frequency plot of (a) the original ES contaminated EMG signal (Repetition 1) and (b) EMD-notch filtered signal using EMD-HT. Each mode is plotted with a different color, depending on its energy in the signal (color bar on right).

all 35 Hz and its harmonics from the EMG signal retaining 20–150 Hz of EMG data (Fig. 6). However, within 20–150 Hz band, Notch filtering did block the 35 Hz and its two harmonic frequencies (70 Hz and 105 Hz) of the EMG data.

A. Time Frequency Representation Using EMD-HT

Time-frequency representations of extracted IMFs can provide a tool for visual separation of stimulus artifact from the EMG data and also show how the instantaneous frequencies of artifact and EMG converge or diverge with respect to time. Fig. 7 shows the time-frequency representation of the test

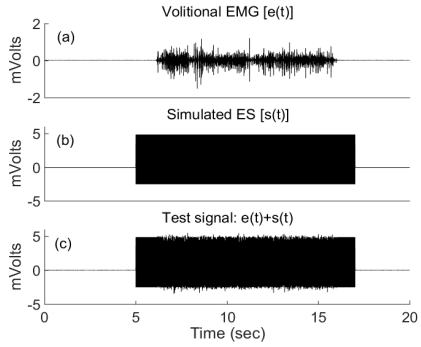

Fig. 8. Simulated ES contaminated EMG construction for validation of EMD-Notch filtering, (a) volitional clean EMG, (b) simulated ES artifact generated (Ratio=4) and (c) addition of volitional clean EMG and simulated ES artifact.

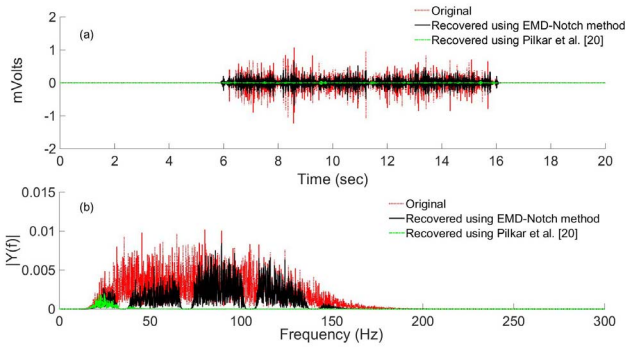

Fig. 9. (a) original volitional clean EMG (red) and the recovered EMG using EMD-Notch algorithm (black) and (b) their FFT representations, respectively.

signal [Fig. 7(a)] and the EMD-Notch filtered signal [Fig. 7(b)] for the Hilbert transforms of each IMF plotted together.

The time-frequency representation of the signal shown in Fig. 7 illustrates how the stimulus artifact engulfs the EMG activity. The IMFs containing the stimulus frequency (35 Hz) and its harmonics are represented by "high-energy" colors [Fig. 7(a)]. Further, the EMG onsets show drops or shifts in frequencies in the time-frequency plot (marked by arrows in Fig. 7(b) subplot).

B. Performance Evaluation Using Simulated ES and Volitional EMG

Fig. 9 shows the outcome of EMD-Notch algorithm and compares EMG signal recovered from the simulated ES [Fig. 8(c)] with the original EMG signal [Fig. 8(a)]. The algorithm was able to extract the volitional EMG signal with minimal loss of data. The recovered EMG is matched with the original signal with respect to amplitude and timing. Furthermore, the FFT comparison [Fig. 9(b)] shows that only the data at 35 Hz and its three harmonics were blocked due to Notch filtering, while spectral content between 20–150 Hz was recovered with minimal loss. EMD-Notch algorithm recovers much wider spectral components of EMG data compared to the signal recovered using our previous method [20] which is shown in green in Fig. 9. Although the time domain profile of the signal recovered using only basic EMD seemed to retain the shape of underlying EMG activations [20], EMD-Notch

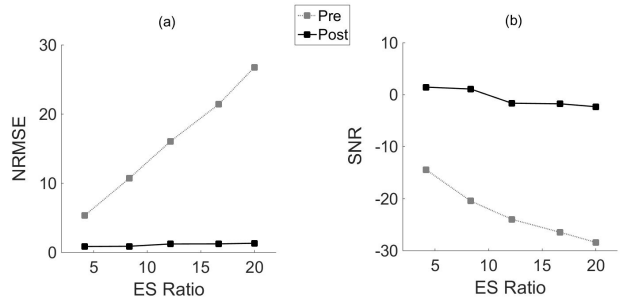

Fig. 10. (a) Normalized root-mean-squared error (NRMSE) and (b) signal to noise ratio (SNR) between the unstimulated volitional EMG and filtered EMG at different levels of ES amplitude, pre (dashed) and post (solid) EMD-Notch filtering.

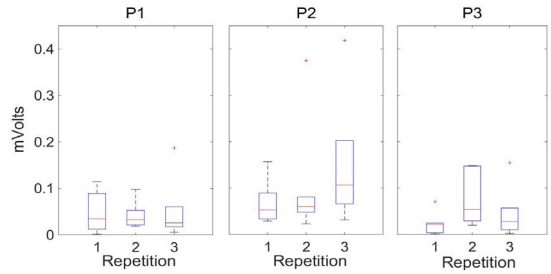

Fig. 11. (a) Comparison of EMD-Notch filtered EMG RMS amplitudes during P1: ES start to ES max (ramping up); P2: ES plateau; P3: ES max to zero (ramping down) for each repetition.

filtering yields the signal with much higher amplitudes that match closely with the volitional EMG signal [Fig. 9(a)].

The magnitude of the EMG spectral distribution recovered from the simulated data also match closely indicating that there was no distortion introduced by application of EMD-Notch filter. However, as mentioned previously, some of the frequency components were lost during Notch filtering.

NRMSE values before the EMD-Notch filtering were higher compared to the same calculated on the filtered signal [Fig. 10(a)], indicating the extent at which the algorithm was able to recover the original volitional clean EMG signal by removing ES artifact. NRMSE values (pre and post) show linear increments as the ratio or the amplitude of ES increased. Importantly, the slope of $NRMSE_{pre}$ was greater than $NRMSE_{post}$ suggesting that ES amplitudes did not affect the performance of the EMD-Notch algorithm. As expected, SNR of the unfiltered signal increased linearly with increasing level of ES [Fig. 10(b)]. Similarly, SNR of the filtered signal was higher than the ES contaminated EMG suggesting the degree of artifact reduction and quality of the output signal.

C. Amplitude Analysis of Able Bodied EMG Data

EMGs collected from the six AB individuals for each of the three repetitions within the ES trial were processed using the EMD-Notch algorithm. The RMS amplitudes of recovered EMGs (Fig. 11) for each repetition were compared using repeated measures ANOVA to show no significant difference among repetitions in the RMS EMG values ($F(2, 10) = 1.85$, $p = 0.21$). Also, ANOVA showed no significant difference in

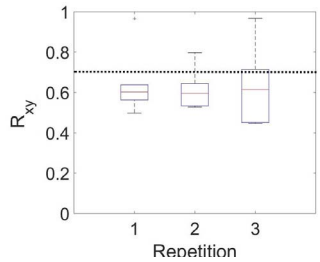

Fig. 12. Cross correlation (R_{xy}) values between force and RMS EMG during each repetition. Dotted line indicates R_{xy} between the clean EMG and force collected during a maximum voluntary contraction (MVC) trial from an AB individual.

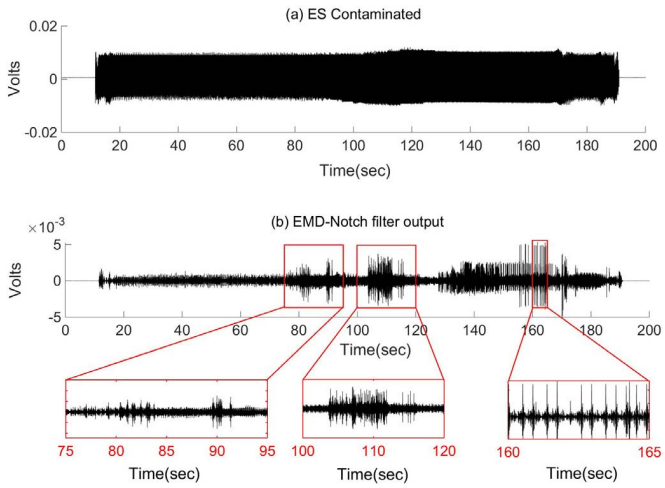

Fig. 13. (a) ES contaminated EMG signal and (b) EMG-Notch filtered output for LRF muscle of a 26-year-old male with incomplete SCI. For better readability, three windows of filtered output are expanded.

the mean EMG amplitudes during P1, P2 and P3 within each repetition of ES ($F(1.01, 5.07) = 1.05$, p=0.35).

The force produced by electrically stimulated contractions was measured using a 6 degrees of freedom load cell (AMTI, AMTI. Watertown, MA, USA). The EMD-Notch algorithm was further validated by comparing the RMS (50 ms) profiles of artifact free EMG with the force profiles during each repetition using the method of cross-correlation. Cross-correlation values computed for all the AB subjects are presented in Fig. 12.

The box plot shows that the median value of cross correlation index (R_{xy}) within each repetition was greater than 0.5, suggesting that the ES induced EMG activation profiles were correlated to the resultant force profiles produced. In addition, repeated measures ANOVA showed that there was no significant difference across all repetitions for the mean cross-correlations ($F(2, 10)$ =0.189, p=0.831).

D. Application of EMD-Notch Algorithm on SCI EMG Data

In order to assess the applicability or potential feasibility of the EMD-Notch filter algorithm (which works well in AB data), we also applied it to the EMG data collected from an individual with SCI. Fig. 13 shows one repetition of ES on an individual with SCI and the EMD-Notch filtered output.

The algorithm was successful in extracting the suppressed bursts of muscle activations during ES.

IV. Discussion

Use of surface electromyography (EMG) can potentially serve as a non-invasive way to study the muscle response to ES; however the separation of EMGs from ES artifact has been difficult to achieve [14]–[16] due to the amplitude and frequency overlap of the stimulus artifact and EMG response. The significance of this study is the development and validation of a robust signal processing algorithm to extract EMG during FES applications to study the physiological response of stimulated muscles for six able bodied controls. The algorithm provides an avenue to study the response of the stimulated muscle to the stimulation parameters, and potentially provides a means for optimizing ES parameters to develop efficacious interventional clinical programs to achieve desirable neuromuscular and functional outcomes.

While hardware approaches to extract the stimulus during the data collection have been previously suggested [14], [16], we propose that a software algorithm will be more advantageous because it enables the processing of pre-collected offline data. Also, with the addition of any Matlab or LabView driven data acquisition platforms, the software would be able to clean the data instantaneously, without making any changes in the data acquisition hardware.

As previously mentioned, O'Keeffe et al developed a two stage peak-detection algorithm for isolation of EMG from stimulus artifact however the method is only effective when the M-wave and the artifact spike do not overlap [15]. By comparison, the basic EMD method outlined in this paper can distinguish the different modes that constitute the EMG signal at any instance of time using method of sifting. Thus the EMD method can provide a more optimal platform for extraction of M-wave from stimulus spikes. Heffer et al. removed the stimulus artifact by linear interpolation using neighboring sample points to replace the stimulus artifact [17]. Their technique is dependent on identification of the artifact event and the artifact duration must remain less than both the inter-stimulus interval and the time course of the action potential [17]. By comparison, the EMD based decompositions are simple to perform and do not need prior knowledge about signal properties such as artifact durations or instances for extracting the EMG as they are completely data driven.

A. Significance of EMD Based Approach

From a physiological viewpoint, the EMD algorithms preserve the stochastic nature of the nonstationary EMG signal whereas in the more traditional decompositions performed using a FFT or a wavelet, the signal is broken down in waves (sine for FFT) that have little to do with the physiological processes of muscle contraction [22]. Recently, Yochum et al. developed a wavelet based decomposition algorithm to extract ES artifact from EMG [37]. However, EMG comparison pre and post decomposition in frequency domain is not presented which makes it difficult to evaluate the quality of the filtered signal. Further, how the algorithm

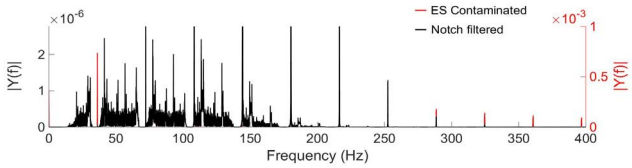

Fig. 14. Original ES contaminated EMG (red) and the recovered EMG using only notch filtering (black).

will perform during electrically stimulated dynamic movement (ES with volitional contraction) is not clear. Instead of trying to distinguish between the artifact and EMG frequencies and then filtering them, EMD algorithms separate the signal into the artifact and EMG components in a way that preserves the intrinsic properties of each. Such separation of artifact and EMG components from the ES contaminated signal is driven by a fact that EMD acts as a filter bank [22], [34]. It is this special characteristic of the decomposition that makes it possible to distribute the ES artifact components in first 4 or 5 of the IMFs. This further allows us to apply a simple statistical condition of standard deviation to automate the classification of artifact IMFs and EMG IMFs.

In our preliminary investigation reported earlier [20], EMD was able to isolate the volitional EMG bursts during ES, though some of the higher frequency content (>50 Hz) was still lost (Fig. 9). This presents a significant challenge when working with the EMG data collected from stimulated muscles from neurologically impaired populations. It is essential not to lose any activity in the signal because there may only be minimal present. It is also desirable not to identify any false activity (stimulus harmonics) as a muscle response to the stimulus. Using the EMD-Notch approach, we were able to recover the data up to 150 Hz which was the upper limit for our pre bandpass filtering. With this knowledge, one would be interested in using only notch filtering approach to filter out all the ES fundamental frequency and its harmonics. However, due to the overpowering nature of ES artifact, a series of notch filters suppresses artifact components partially and also reveals the secondary or tertiary ES harmonics of lower intensities which still dominate the pure EMG activity (Fig. 14) and require further filtering, eventually resulting in loss of significant data.

Our validation process using the unstimulated volitional EMG and simulated ES showed that the EMD-Notch algorithm was successful in recovering the signal (Fig. 9). Reduced NRMSE values post EMD-Notch filtering indicated the accuracy of the extracted output from the ES contaminated signal. Better SNR post EMD-Notch filtering suggest the quality and robustness of filtering operation.

EMG-force relationship has been previously established [38]. The cross correlation analysis of EMD-Notch filtered RF signal and the resultant force measured at the knee joint can provide a secondary validation for the filtered data, as the RF EMG should be correlated with the generated force. The cross correlation values (R_{xy}) of filtered EMG and the force measured during ES induced activation were acceptable (>0.5), supporting the use of EMD-Notch filtering since the artifact removed EMG signal follows the EMG-Force relationship without showing any irregularities, transformations or additions to the signal as a result of the EMD-Notch filtering. Note that the cross correlation value calculated using the clean EMG and force measured during a MVC trial was 0.71 (Fig. 12) which was within the range of cross-correlation values for all AB subjects during all repetitions. However, it should also be noted that the resultant force generated is not just the result of a single isolated muscle but a combinatory effect of work done by multiple muscles.

We also showed the application of EMD-Notch filtering on the EMG collected from an individual with SCI and successfully extracted the muscle activations during FES. The EMG amplitude of the extracted activation from an individual with SCI was in the range of ±0.5 mV which was much greater than the extracted activation amplitudes for the AB individuals. The differences in extracted activation amplitudes reflect the differences in the amplitude of ES_{max} for ES induced contraction. For the individual with SCI ES_{max} was 91 mA compared to much lower mean ES_{max} amplitudes (43± 9.45 mA) for all the AB individuals (Table I).

B. Current Limitations

One limitation of our approach is that the basic EMD algorithm is suffered by the problem of mode mixing. Mode mixing is characterized by the IMFs consisting of oscillations of dramatically disparate scales or a signal of a similar scale residing in different IMF components [39]. This necessitated the need for notch filtering of such frequency overlapping IMFs in IMF_{art} to isolate the ES artifact from EMG activity of pure muscle contractions. As a result, the EMG signal components at notch filter frequencies (harmonics of ES artifact) were lost. Although the solution for mode mixing problem has already been established in terms of noise assisted ensemble EMD (EEMD) [39], optimization of EEMD (amount of noise to be added) for electrically induced EMG needs further investigation for meaningful interpretation of the decomposed IMFs. Future work could also possibly evaluate replacing the notch filter operation by combining EMD with Independent component analysis (ICA), another computational method for separating a multivariate signal into subcomponents to isolate ES from actual muscle activations during ES, especially during multi-channel FES. The method would enhance the further separation of high frequency ES harmonics from EMG data based on the statistical properties of the data. Further validation could be investigated by measuring the sensitivity of the algorithm at different levels of efforts (or %MVCs) for able bodied individuals and comparing force-EMG correlations. Further, with a larger sample size, MVC amplitude normalization would allow for a within-subject and within-group comparisons.

EMG Sampling Recommendation During FES Applications: One of the major consideration for analyzing EMG data during ES is the sampling rate of the data acquisition systems. In this study, EMG data was collected at 2520 Hz with channel bandwidth set to maximum 500 Hz. Although this set up

is sufficient to record actual muscle responses (<500 Hz), it might not be sufficient enough to recover a train of ES pulses of 300μs pulse width with repetition frequency of 35 Hz. Such sampling rates and bandwidths could result in widened ES pulses (pulse width > 300 μs) due to aliasing and ringing artifacts. Hence, using higher sampling rates ($\approx$ 10 kHz) and wider bandwidth (>2 kHz) are recommended.

C. Potential Clinical Application

Although accompanied by some limitations, our approach shows the potential to assess artifact-free EMG collected during ES contractions and to observe the changes in muscle activation due to change in ES parameters. This EMG can provide significant information about the active motor units for each ES pattern. The intrinsic properties of muscle fibers decide the shape and conduction velocity of motor unit action potentials (MUAPs) which contribute to the spectral characteristics of EMG [40]. Therefore, spectral properties of artifact-free EMG collected during varied patterns of ES can enable us to know the effect of certain frequencies, amplitude, and pulse width of ES on motor recruitment patterns, muscle fatigue and force of intrinsic muscle contraction.

V. Conclusion and Future Work

We hypothesized that EMD provides a suitable platform for isolating the non-stationary EMG signal from ES artifact by decomposing the signal into physically meaningful IMFs. Basic EMD combined with Notch filtering produced artifact free EMG signal during ES induced muscle activations. Although, some data at certain harmonic frequencies was lost during the reconstruction process, the EMD-Notch approach showed accuracy and robustness to recover EMG at different levels of ES intensities. Overall, the EMD-Notch approach provides an effective, adaptive, fast, easy-to-implement approach to interpret the EMG data during ES.

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